Lesson 8: Applying Masks using Face Detection

Read in an image that contains a face

The `url` below links to an image of actor Adam West on his IMDB page. Or, feel free to use images from the Pixels to Pictures Image Library or your own images. If you use your own images, just make sure to provide the full path to the image so MATLAB can find it.

```
url = "https://m.media-amazon.com/images/M/" + ...
    "MV5BNTY4MjMwNjc0OF5BMl5BanBnXkFtZTgwMjA2NDg1MjI@._V1_QL75_UX219_.jpg";
I = imread(url);
```

View input image with Face Detection

Use a Computer Vision cascade object detector to detect faces.

```
faceDetector = vision.CascadeObjectDetector();
bbox = faceDetector(I);  % bounding box: [x y height width]
% x/y corresponds to column/row location of upper left corner of box

% Add bounding box and display result
J = insertShape(I,"rectangle",bbox,LineWidth=3);
figure
imshow(J)
```

Load mask images

The code below uses the beard and Batman masks. Feel free to try others! Depending on the masks you choose, you may need to adjust the mask scale and location bias terms below to get a good fit on the detected face.

```
load("masks.mat")
mask1 = beard;
mask2 = batman;

figure
imshowpair(mask1,mask2,"montage")
```

Adjust masks

```
x = 1.1;  % scale mask width as percentage of face width
scale1 = x*bbox(4)/size(mask1,2);  % scale mask columns by face width
scale2 = x*bbox(4)/size(mask2,2);  % scale mask columns by face width
mask1 = imresize(mask1,scale1);
mask2 = imresize(mask2,scale2);
```

Set location where mask should be overlayed

```
dx = (1-x)/2;       % shift to center left/right on face
bias1 = [ 0.5 dx];  % shift down and use dx to center left/right
bias2 = [-0.5 dx];  % shift up and use dx to center left/right

% Shift mask by percentage of the face height/width
loc1 = bbox([2 1]) + bias1.*bbox(3:4); % location: [row column]
loc2 = bbox([2 1]) + bias2.*bbox(3:4); % location: [row column]
```

Superimpose the mask on the face image

```
final = superImpose(I,mask1,loc1);
final = superImpose(final,mask2,loc2);
```

View final image

```
figure
imshow(final)
```